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Traffic Analysis and Visualisation from Sparse Mobility Data

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Abstract
Mobility is a fundamental service in Smart Cities, as commuting between different urban areas underpins daily life. Accurate visualisation of mobility data benefits not only commuters but also traffic managers and city authorities, enabling the identification of congestion hotspots and dominant traffic flows, and supporting timely, data-driven decisions. We propose a novel framework that integrates meshless approximation using Radial Basis Functions with Laplacian-based graph signal recovery to reconstruct and visualise spatio-temporal traffic data across urban road networks. This approach yields smooth, continuous estimates of speed and volume, even in data-sparse regions, and enables the identification of high- and low-traffic zones, as well as dominant intra-city flows. We further estimate vehicular flux using a discrete formulation of Fick's diffusion law, which reveals patterns of accumulation and dispersal. A road segmentation strategy enhances spatial resolution by dividing roads into smaller segments, enabling finer interpolation and eliminating artefacts in non-road areas. Our findings also reveal areas where traffic volume and speed are negatively correlated. Overall, our framework combines robust data modelling, estimation, and visualisation techniques to support effective urban mobility planning and decision-making.

Keywords Traffic data analysis and visualisation · Graph signal recovery · RBF interpolation · Fick's diffusion law · Urban network · Vehicular trajectories

1 Introduction

The generation of mobility and traffic data sets of distinct types (e.g vehicles' or people's positions, speeds, or trajectories) is rapidly increasing as different collection methods are being developed, particularly in Smart Cities [27]. Data visualisation is gaining importance in other fields as it represents a tool to obtain insights from complex data sets. Transportation and mobility require various techniques to leverage the vast amount of acquired information. Even if urban data sets consist of large amounts of information, they may be available only in a limited part of the city for various reasons, such as privacy concerns or implementation costs. Interpolation and extrapolation methods for structured [9] and non-structured data [22] represent a solution for extending real-world datasets, such as those in the mobility and traffic domains, enabling the availability of valuable information across the entire urban network of a city.

The main types of traffic data include (i) vehicle movements with data on vehicle speeds, locations, and types (e.g., cars, buses, motorcycles), (ii) congestion levels with information on traffic density, and slow-moving or idle vehicles, including locations of traffic incidents, accidents, or road hazards, (iii) traffic flow representing the direction and volume of traffic across different routes; and (iv) environmental impact with information on emission levels based on traffic conditions and vehicle types. *Information visualisation* of traffic data refers to transforming raw traffic data into graphical representations, allowing users to easily understand and analyse urban mobility patterns, trends, and dynamics. This representation is useful in identifying traffic bottlenecks or interruptions within an urban grid. Through time series, dynamic changes in traffic patterns are applied to show how congestion builds and dissipates.

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This work proposes an integrated methodology for the reconstruction and visualisation of spatio-temporal traffic data across urban road networks, combining meshless approximation using *Radial Basis Functions* (RBFs) [23] and Laplacian-based graph signal recovery techniques [8] (Sections 2, 3). The proposed approach enables the generation of smooth, continuous representations of vehicular volume and speed, even in areas with sparse data, and supports comprehensive two-dimensional visualisations of traffic dynamics at specific time intervals. To this end, we review prior research on signal recovery for both Euclidean and non-Euclidean domains, focusing on methods particularly relevant to traffic modelling and mobility analysis—namely, RBF interpolation, graph signal processing, and the application of discrete Fick's diffusion law for modelling vehicular flux.

To overcome data sparsity and improve network-wide coverage, we employ a graph-based modelling approach in which roads are modelled as nodes and intersections as edges. Through graph Laplacian-based signal recovery, we estimate traffic conditions across unmeasured segments, thus enabling more complete and accurate urban traffic visualisation. RBF interpolation plays a central role in this process, providing smooth approximations of traffic variables such as speed, volume, and flux. The use of different RBF kernels (e.g., Gaussian, inverse multiquadric) allows flexibility in adapting to data density and spatial characteristics. The visualisation framework integrates multiple spatial representations, including point-wise plots, heat maps, contour maps, vector fields, and diverging maps, offering a rich, multidimensional perspective on traffic behaviour. Furthermore, we introduce advanced traffic metrics by estimating vehicular flux using a discrete formulation of Fick's diffusion law, allowing the identification of accumulation and dispersal zones across the road network. To improve the spatial granularity of the visualisations, we propose a road segmentation methodology that divides roads into smaller linear segments. This choice allows for finer interpolation of traffic variables, resolves visual ambiguities in non-road areas, and enhances interpretability across multiple levels of spatial detail. Together, these components form a comprehensive framework for traffic data reconstruction and visualisation, supporting both high-level overviews and detailed local analyses of urban mobility systems.

We apply the methodology to real-world traffic data from the PNeuma experiment [2], which includes GPS trajectories of over 10,000 vehicles—cars, motorcycles, taxis, and buses—collected in the central business district of Athens. This dataset facilitates the analysis of traffic patterns over time and by vehicle type, offering a dynamic and realistic evaluation environment (Sections 4, 5).

2 Related Work

We review previous work on visual representations in the mobility and transportation domains, as well as distinct approaches that have been tested with various types of data. Moreover, we revise methods to complete missing data on Euclidean and non-Euclidean structures used to store mobility and traffic information, which serve as tools to overcome the lack of information due to unfavourable circumstances (e.g. technical errors, implementation costs, or privacy concerns) that could diminish the output of a visual representation. Furthermore, we revise the discrete version of Fick's diffusion law, which supplies a mechanism to handle flow data in graph-based structures. Examinations, reviews, and surveys of technologies and tools, including data visualisation, for dealing with large amounts of information in transportation and mobility systems have been addressed in [24]. The importance of data visualisation [1] relies on the improvement of intelligent transportation systems through the combination of visual interfaces with the analysis from experts in the mobility and transportation communities. In [12], visual analytics incorporating graph structures at the street and sub-region levels have been evaluated with taxi trajectories of Shenzhen, China, leveraging the use of centrality metrics to reveal traffic patterns and important streets. The use of graph structures is favourable in mobility and traffic models as it provides several theoretical tools that can enhance representations of unstructured data, and also the inclusion of properties that are intrinsic in graph structures (e.g., node centrality metrics, and weighted edges). Another example is the creation of virtual links for predictions of Origin-Destination flows and flow-capacity levels, which in conjunction with neural networks, has been used to complete missing Origin-Destination data [14]. Novel frameworks for day-to-day evolution of departing time and mode choices in transportation systems based on Origin-Destination data have been developed, as they model users decisions based on historical data [15].

Visual analytics has also been combined with Machine Learning to represent traffic predictions from GPS data in New York City [17]. Furthermore, a data visualisation pipeline for the development of tools aiming to support a transport analyst in decision-making has been proposed and applied to a user scenario on a railway line of the Paris metropolitan area [25], remarking that such a tool can be generalised to road and air traffic data. Graphs constructed from the urban network have also been leveraged as an auxiliary framework to identify main flows within the roads of a city. For instance, using the *primal graph* approach (i.e., representing the roads as edges, and road intersections as nodes) in combination with spatial interpolation methods, numerical methods have been utilised to estimate the direction of the vehicular flow under given density assumptions and grid partitions [21]. Nevertheless, methods using actual instant data and not relying strongly on an explicit grid still need to be explored.

Radial Basis Functions Interpolation RBF interpolation approximates a function in an Euclidean domain when only some sampled values at scattered and unordered points are known. A function $\Phi : \mathbb{R}^d \longrightarrow \mathbb{R}$ is called *radial* if there exists a *basis function* $\varphi : [0, \infty) \longrightarrow \mathbb{R}$ such that $\Phi(\boldsymbol{z}) = \varphi(r)$ where $r = \|\boldsymbol{z}\|$ [9]. Approximating a real-valued function *f* at a point $\boldsymbol{z} \in \mathbb{R}^d$ using *M* radial basis functions consists of expressing it as $f(\boldsymbol{z}) = \sum_{j=1}^{M} c_j \Phi_j(\boldsymbol{z})$ for some $\boldsymbol{c} = (c_1, \ldots, c_M) \in \mathbb{R}^M$, where each Φ_j is a radial function centered at $\boldsymbol{\xi}_j \in \mathbb{R}^d$, i.e. $\Phi_j(\boldsymbol{z}) = \varphi(\|\boldsymbol{z} - \boldsymbol{\xi}_j\|)$. When the value of *f* is known at *N* points, we have $h_i = f(\boldsymbol{x}_i) = \sum_{j=1}^{M} c_j \phi_{ij}$ for $i = 1, \ldots, N$, where $\phi_{ij} = \varphi(\|\boldsymbol{x}_i - \boldsymbol{\xi}_j\|)$. Then, the coefficients vector $\boldsymbol{c}$ can be found by solving a linear system $\boldsymbol{A}\boldsymbol{c} = \boldsymbol{h}$, $A_{ij} = \phi_{ij}$, with *M* variables and *N* equations [16]. Typical global radial basic functions are Gaussian $\exp\left(-\varepsilon^2 r^2\right)$, inverse quadratic $\left(1 + \varepsilon^2 r^2\right)^{-1}$, and inverse multiquadric $(\sqrt{\varepsilon^2 r^2})^{-1/2}$.

Regularisation-based Recovery of Graph Signals Several graph-based methods [4, 5, 22] for recovering the values of a graph signal from the known values on a small subset of nodes are based on the trade-off between the precision of the recovered values on the known nodes (i.e. interpolated value), and the similarity of values between pairs of connected nodes (i.e. signal graph smoothness). For undirected graphs, the *Laplacian quadratic form* $\boldsymbol{x}^\top \boldsymbol{L}\boldsymbol{x}$ provides a smoothness measure of a graph signal, where $\boldsymbol{L}$ is the combinatorial Laplacian matrix of the graph. Lower values of the Laplacian quadratic form stand for smoother signals. The Laplacian-based recovery method [8] estimates the complete signal $\boldsymbol{x}^*$ from a sample $\boldsymbol{y}_\mathcal{S}$ known on a subset $\mathcal{S}$ with *m* nodes, by solving the optimisation problem $\boldsymbol{x}^* = \underset{\boldsymbol{x} \in \mathbb{R}^N}{\arg\min} \|\boldsymbol{y}_\mathcal{S} - \boldsymbol{C}\boldsymbol{x}\|_2^2 + \lambda \boldsymbol{x}^T \boldsymbol{L}\boldsymbol{x}$, where $\boldsymbol{C} \in \mathbb{R}^{m \times N}$ is the sampling operator that samples the nodes on $\mathcal{S}$, i.e. being $\mathcal{S} = (v_{\pi(1)}, \ldots, v_{\pi(m)})$ then $C_{ij} = 1$ if $j = \pi(i)$ and 0, otherwise. Using first-order optimisation criteria, the optimal points $\boldsymbol{x}^*$ of the objective function is the solution to the linear system $(\boldsymbol{C}^\top \boldsymbol{C} + \lambda \boldsymbol{L})\boldsymbol{x}^* = \boldsymbol{C}^\top \boldsymbol{y}_\mathcal{S}$. The parameter $\lambda > 0$ represents the trade-off between the error on the sampled nodes and the smoothness of the recovered signal. When λ is close to zero, the error on the sampled nodes is small, as the Laplacian quadratic form can be considerably large without affecting the minimisation problem. In contrast, when λ is large, the optimisation problem prioritises minimising the Laplacian quadratic form and, consequently, the variability of the values on the sampled nodes will be larger.

Discrete Fick's Diffusion Law The Fick's law of diffusion of a flow $\boldsymbol{q}$ on an Euclidean domain of a substance with concentration $\boldsymbol{\phi}$ states $\boldsymbol{q} = -k\nabla\boldsymbol{\phi}$, i.e. the flux flows from regions of high concentration to regions of low concentration, with a magnitude proportional to the concentration gradient and coefficient of diffusivity *k*. On a graph structure, an algebraic manipulation [7] of Fick's law leads to its discrete version $\boldsymbol{q} = -k\boldsymbol{L}\boldsymbol{\phi}$, where $\boldsymbol{L}$ is the combinatorial Laplacian of the associated graph. Moreover, an estimation of the concentration vector can be obtained through $\widehat{\boldsymbol{\phi}} = -\boldsymbol{L}^\dagger \boldsymbol{q}$, where $\boldsymbol{L}^\dagger$ is the pseudo-inverse matrix of the Laplacian matrix.

3 Approximation and Visualisation of Traffic Data

Spatio-temporal traffic data in the form of GPS trajectories brings out distinct ways to visualise them. We start enumerating visualisation techniques that we will use to leverage the different variables of our traffic data set (Section 3.1). Then, we use point-wise representations using vehicles' positions and apply heat maps according to their instant speeds at that position (Section 3.2). Since the collected data is available only on a limited number of roads, we propose global-scale visualisations using a graph signal recovery process and RBF interpolation functions (Section 3.3). Then, to address granularity within the road network, we propose a road segmentation methodology (Section 3.4) that provides visualisations with higher levels of detail, even on those roads where data is not available.

3.1 Visualisation Tools

Assigning colours to scalar values through colour maps permits the identification of the distribution via different types of visualisations. For instance, colour-coded segments of roads, where fast-moving traffic is light yellow, moderate speeds are purple, and slow or stationary traffic is black, show the variation of vehicle speeds or volume across a road network. Moreover, visualising traffic data with RBF interpolation will produce smooth, continuous maps that help decision-makers and analysts identify patterns in traffic conditions across the entire network. To this end, we will apply

– *uniform heat maps with point-wise representations* using the vehicles' position and piece-wise linear road segments. Vehicles with higher speeds and roads with larger average speeds are in light yellow, while the slowest ones are in black. Furthermore, *uniform heat maps from RBF interpolation* will be applied to show variations in traffic variables like speed or volume using colour gradients. Areas with high traffic speeds or volume are in light yellow, while congested or slow-moving areas appear in black;

– *diverging heat maps* with RBF interpolation and road representations for colour-coding negative and positive values of vehicular flux. Zones with red colour stand for positive values with more incoming vehicles than outgoing ones, and zones with blue colour stand for negative values where the number of outgoing vehicles is larger;
– *contour maps*, where contour lines drawn over RBF-interpolated traffic data represent zones of equal traffic speed or volume. This visualisation helps to identify critical areas where traffic conditions change rapidly (e.g., from free-flowing to congested);
– *vector field visualisation*, where RBF interpolation of traffic flow data is visualised using arrows or vectors to represent the direction and magnitude of traffic movement and to identify the flow of vehicles across complex intersections or road networks.

We apply RBFs to traffic data approximation and visualisation to model and interpolate traffic patterns. RBFs allow for smooth interpolation of traffic-related variables (e.g., speed, volume, or flow) based on known data points, allowing the creation of continuous representations of traffic conditions across a road network. More precisely, we apply RBF approximation to estimate traffic speed, volume, and flow across a road network, even where no direct measurements are available. RBF interpolation for traffic data approximation and visualisation enables the smooth and continuous representation of traffic variables like speed, volume, and flow, even with sparse or unevenly distributed data. Using data from sensors or GPS devices on vehicles at specific points, RBFs generate a smooth field representing the traffic speed across the entire network and work well even when data points are irregularly spaced, a common situation in traffic data (e.g., when sensors are placed unevenly across a city). Different RBF kernels (e.g. Gaussian, inverse quadratic, inverse multiquadric) can be chosen according to the characteristics of the traffic data being modelled, providing flexibility in approximation. Finally, RBFs can model complex traffic dynamics, including rapid changes in speed or volume, making it a powerful tool for traffic management. Providing accurate and detailed visualisations, RBFs help to improve the analysis of traffic flow, congestion, and urban mobility.

3.2 Point-wise Representations

For our implementations, we use the real-world traffic data set obtained from the *PNeuma experiment* [2], consisting of GPS trajectories of distinct types of vehicles (e.g. cars, motorcycles, buses) collected during different time intervals over the central business district of Athens. The PNeuma data set also includes the instant speeds of each trajectory point computed from its previous positions and time entries. The experiment was performed in such a way as to obtain traffic information for each of the weekdays. We use data corresponding to the time window between 08:30 AM and 09:00 AM on Wednesday, October 24th 2018, which includes the trajectories of more than 10000 vehicles.

For a fixed time, the information is available only on a sub-region of Athens' urban network, and it is visualised using scattered points with longitude and latitude as coordinates with a proper geographical projection, if necessary (e.g. Mercator projection). Using vehicle trajectories of different time entries, we visualise their positions along incremental times and discover parts of the urban network where the number of vehicles is increasing, which can lead to traffic congestion and, consequently, a decrease in vehicular speeds. For each vehicle type, we construct a uniform colour map from its minimum and maximum speed values and use it to assign a colour to the points according to how fast they are moving (Fig. 1). Thus, we identify the areas where vehicles are travelling at higher speeds. Moreover, we use three-time entries, each separated by 5min, to see some roads of the urban network where the number of trajectory points around them is growing, which is particularly more noticeable for cars and motorcycles. Sharing the same colour map among each type of vehicle and time entry allows us to make comparisons between them holistically.

3.3 Global Visuals on the Urban Network and Road Segments

Scattered plots of trajectory data on a limited number of roads do not provide interpretations for areas within the urban network where no information is available. We use RBF interpolation with the vehicles' positions to estimate speed values on a square grid over the urban network to obtain values that can be plotted using a heat map. Since interpolation may lead to negative values, we map them to zero to obtain an appropriate transfer function (i.e. colour map) to represent instant speeds. To apply RBF interpolation of the instant speeds at the position of the recorded vehicles, we use a Gaussian function with scaling factor $\varepsilon = 10$, as it promotes smoothness and fast decay to zero for large radius values. Since RBFs are globally defined on $[0, \infty)$, convergence problems could occur when the number of points to be interpolated is large [23], then we use a regularised version of the associated linear system, i.e. $(\boldsymbol{A} + \mu \boldsymbol{I})\boldsymbol{c} = \boldsymbol{h}$ (e.g., $\mu = 0.1$), where $\boldsymbol{I}$ is the identity matrix of the corresponding dimensions. Although a suitable

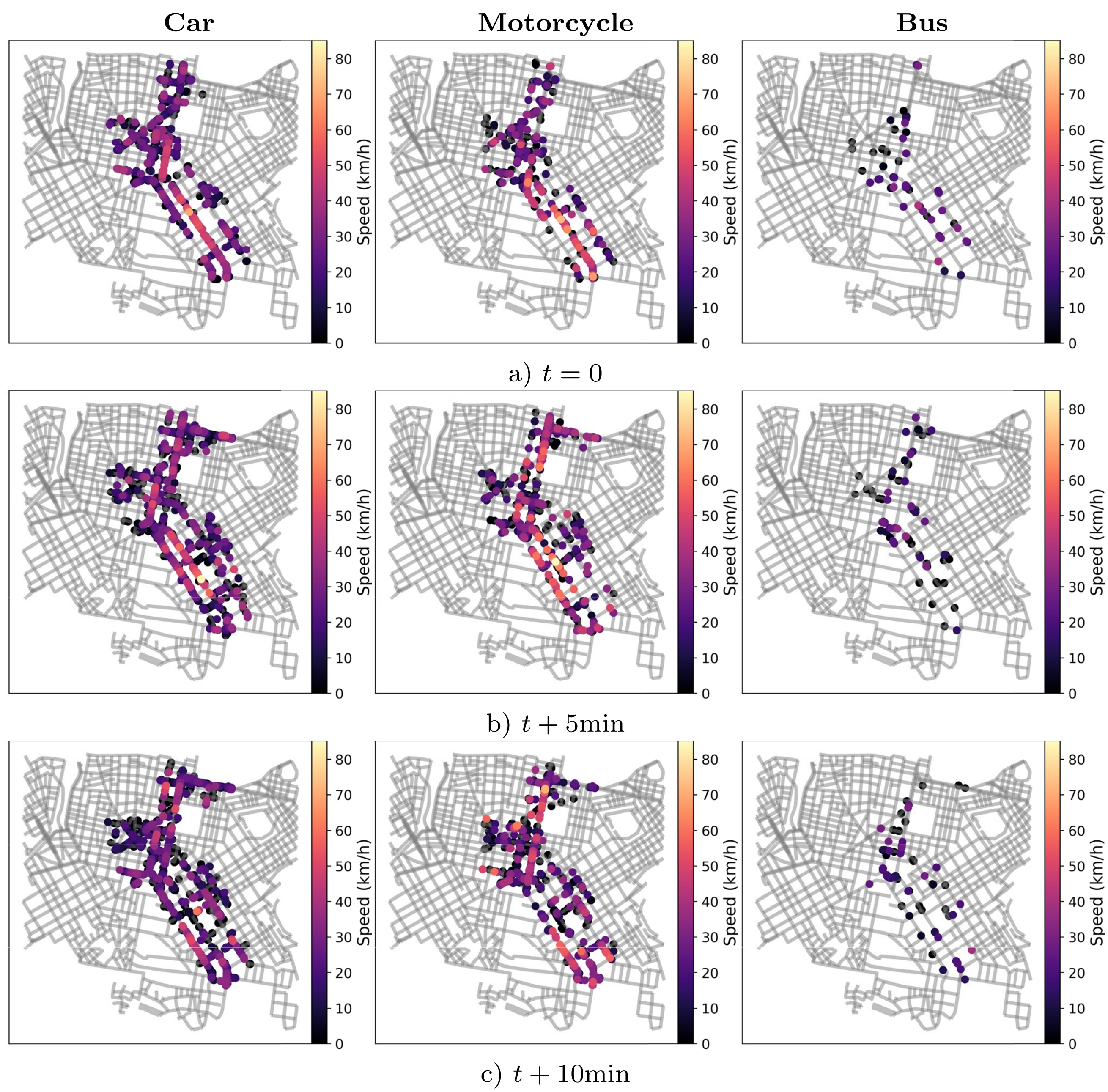

Fig. 1 Trajectories of distinct types of vehicles over the central business district of Athens during three time instants with a 5 minutes' difference between consecutive times. The **GPS points** are colour-coded using a uniform colourmap for the **instant speed** values, where vehicles with higher speeds are shown in light yellow, vehicles with moderate speeds are in purple, and the slowest vehicles are in black. The colour map is shared among each type of vehicle and time instant

choice for the ε and μ parameters is crucial for overcoming modelling issues (e.g., controlling the range of the estimated values, and minimising the empirical error), our purpose in this work is to present a visualisation methodology combining RBF interpolation with a graph signal recovery method. In fact, for the task of identifying critical zones in the urban network, variations of ε and μ show consistent spatial patterns. For an optimal choice of such parameters, we refer the reader to [10, 26]. We interpolate the instant speeds at the GPS positions for 3 types of vehicles: car, motorcycle, and bus, and visualise them using heat maps representations with a shared transfer function (Fig. 2a).

Further traffic information that can be plotted to provide valuable insights for traffic management includes vehicular volume (i.e. number of vehicles) and flux (i.e. difference of outgoing and incoming vehicles). To this end, we construct a road-level graph from Athens' urban network, i.e. its associated *road graph*, where each node represents a road, and

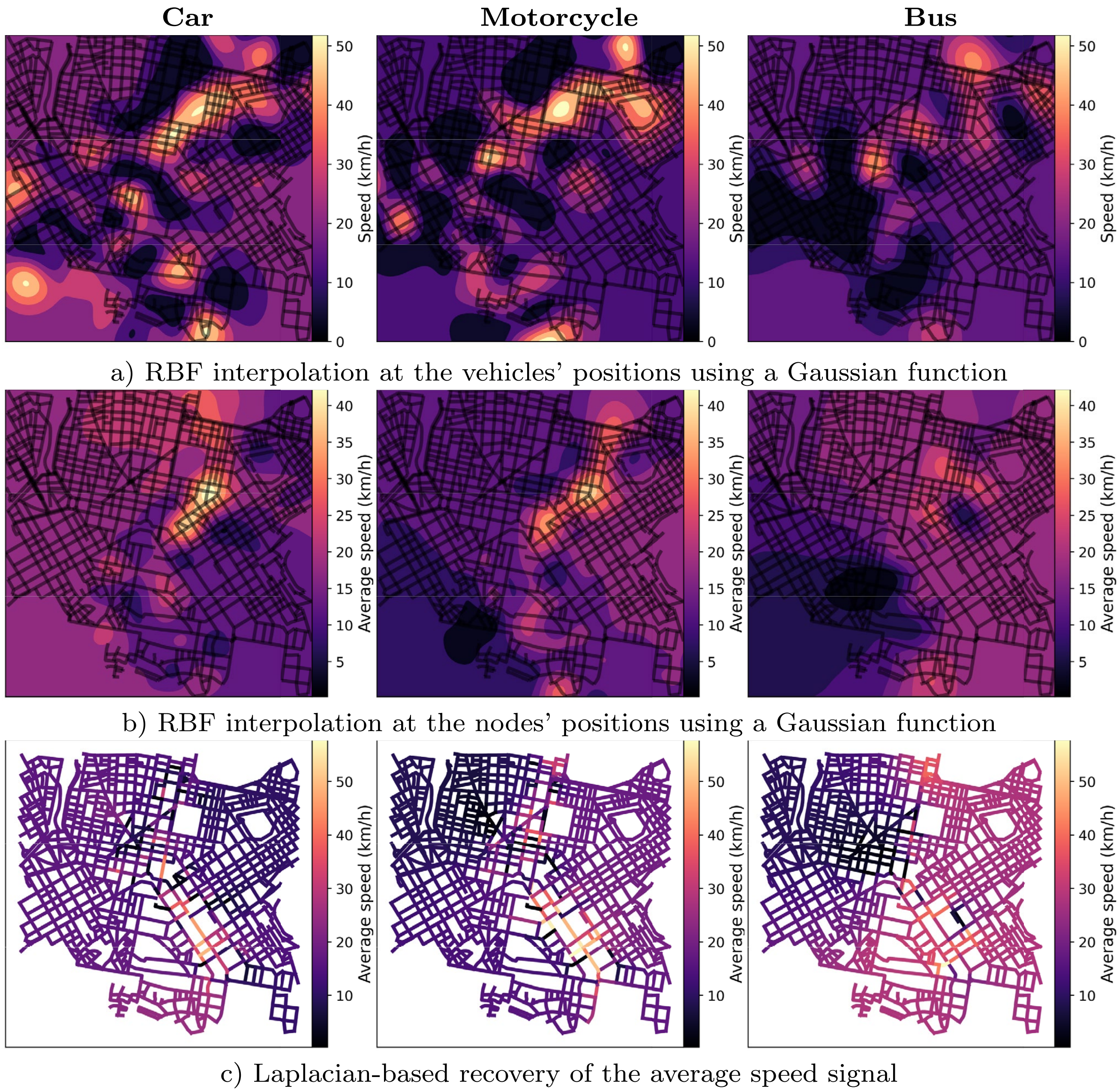

Fig. 2 RBF interpolation of speed values using the **instant speed** at the original vehicles' positions (**a**) and the **recovered average speed** signal at the nodes' embedded positions (**b**), followed by a uniform colour map, to obtain a smooth representation with heat maps over the whole urban network with distinct basic functions. Colour-coded road segments are also used to represent the recovered average speed (**c**), limited to only the road network structure. The colour map is shared among all vehicles

edges represent the intersection between two roads. The concept of a *dual graph* representing a street as a node, in contrast to the approaches involving the *primal graph* with nodes as intersections, has become popular since the development of the *Space Syntax* methodology [11], i.e., a set of techniques for the analysis of spatial networks that has been leveraged by urban designers to understand several structural properties of city spaces. Particularly, the use of the dual approach for our modelling allows us to consider information on an entire road as a consequence of point data along its geometry, as in the case of GPS positions or trajectory data.

In our graph-based model, *graph networks* model roads (nodes) and their intersections (edges) to visualise traffic flow and connectivity. Moreover, we embed nodes of the road graph to $\mathbb{R}^2$ by associating them with the pair (x, y) corresponding to the midpoint of the road segment (e.g. along the geometry). We construct the following graph signals from the PNeuma data set: (i) the *volume* signal on a node is defined as the number of vehicles whose geometrical projection of its coordinates (*x*, *y*) to the road graph belongs to its associated road and (ii) the *average speed* signal on a node is defined as the arithmetic mean of the instant speed (*km*/*h*) of the vehicles projected to the associated road. Since the considered trajectory points from the PNeuma data set correspond to vehicles moving in a single direction on each road, the computation of the average speed is well defined. Road networks have an associated direction by nature; therefore, an alternative to define average speeds in directed graphs needs to be explored in future work, being the use of *multidigraphs* being a possible solution.

The idea of using graph signal recovery before RBF interpolation is, in essence, to have traffic values on each road and then to use the graph embedding to make estimations within the whole urban network. To this end, we apply the Laplacian-based graph signal recovery to reconstruct values on the whole Athens' road graph with regularisation parameter $\lambda = 10$ as larger values of λ guarantee smoother recovered signals. For assessing the precision of the recovered graph signal, distinct approaches can be implemented to measure the error on the known nodes, e.g., through the *normalised maximum error* that measures the proportion of the largest absolut differences on the sampled nodes with respect to the magnitudes of the ground-truth values, as done in a previous work for a wide variety of graph signal recovery methods, including Laplacian-based [19].

Using RBF interpolation at the nodes' embedded positions with a Gaussian function, we estimate the average speed in a squared grid over the whole urban network and visualise it using heat maps with a shared transfer function (Fig. 2b). Similarly, the obtained negative values are mapped to zero. Having reconstructed the average speed to all the roads enables us to colour the roads according to the same transfer function used in the heat maps (Fig. 2c).

To further quantify the performance of the proposed reconstruction pipeline on the PNeuma dataset, we performed a hold-out evaluation and complementary statistical analysis on the recovered signals. Using a 20% random road hold-out repeated over 30 trials, the Laplacian-based recovery combined with RBF interpolation achieved a mean NME of 0.071 ± 0.014 for average speed and 0.084 ± 0.021 for vehicular volume. In contrast, a direct RBF-only interpolation baseline yielded 0.131 (speed) and 0.147 (volume), confirming that graph recovery substantially improves estimation fidelity by enforcing network consistency. We also evaluated the spatial stability of the recovered field under increasing segmentation granularity (4, 8, 16), and 20 partitions per road. The mean absolute variation of recovered speeds along the same road across granularity levels was below 4.6%, demonstrating that the underlying reconstruction is robust with respect to road discretisation and that segmentation primarily increases visual interpretability rather than altering the recovered values.

To quantify the temporal effect, we measured the change in congestion on the subnetwork between $t = 0$ and $t = 10$ minutes, observing an average increase of $+12.3\%$ in vehicular volume on the most congested roads for cars and $+15.7\%$ for motorcycles, in line with visual evidence of accumulation. Flux estimates exhibit a corresponding negative drift of -9.8% (cars) and -12.1% (motorcycles) in the same zones, confirming the formation of bottlenecks. Finally, we computed the Pearson correlation between speed and recovered volume, obtaining -0.64 over the active portion of the network, quantitatively supporting the negative correlation between congestion and mobility that is only qualitatively hinted at in the visualisations. This value increases in magnitude (-0.71) in the central high-density subnetwork, consistent with the fact that congestion effects are spatially localised and nonlinear. Overall, these results show that

(i) graph-based recovery reduces reconstruction error and suppresses spurious values outside road areas;
(ii) the reconstructed field is numerically stable across segmentation levels; and
(iii) temporal evolution is not only visually observable but also measurable through changes in flux imbalance and speed–volume correlation.

This graph-based approach grants us the possibility to obtain further visualisations (e.g. vehicular volume and flux) that were not possible with the point-wise trajectory data. Through the Laplacian-based method, we reconstruct the volume signal at all the nodes of the Athens road graph and use RBF interpolation at the nodes' positions to estimate the vehicular volume in the entire urban network using a Gaussian function (Fig. 3a). Considering the volume of vehicles on the roads as a concentration vector, we apply the discrete Fick's diffusion law with diffusivity coefficient $k = 1$ to find the flux vector $\boldsymbol{q}$ on the Athens' urban network, where each component q_i denotes the flow at the position $\boldsymbol{x}_i$ of node i defined as q_i=number of vehicles leaving $\boldsymbol{x}_i$ minus the number of vehicles entering $\boldsymbol{x}_i$, then negative values represent accumulation of traffic, and positive values relaxation of traffic. Finally, we apply RBF interpolation at the nodes' positions with a Gaussian function (Fig. 3b) to estimate the vehicular flux over the whole of Athens' urban network. Although the use of Fick's diffusion law can quantify vehicular flux in a

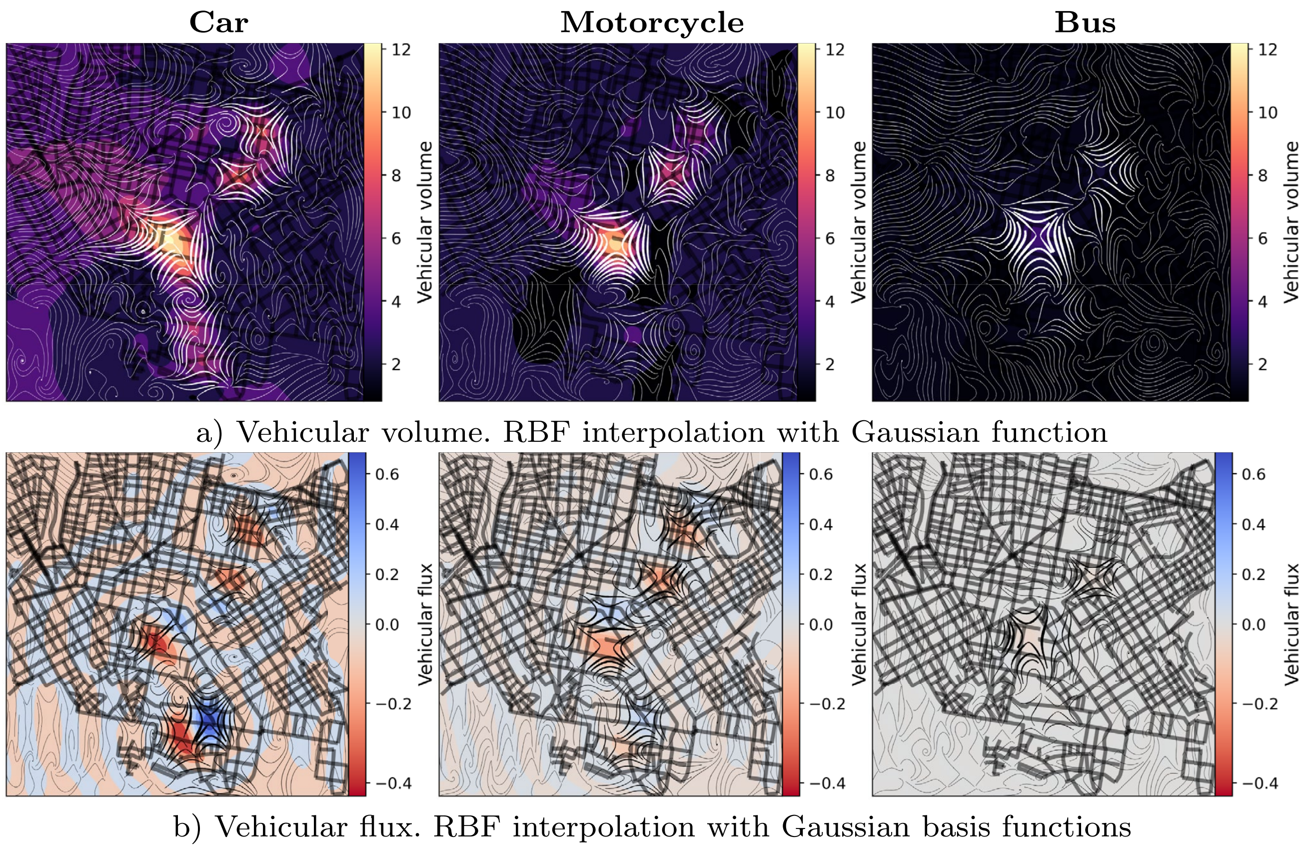

Fig. 3 RBF interpolation using the **recovered volume** at the nodes of the Athens' road graph (**a**) followed by a uniform colour map to obtain a smooth representation with heat maps over the whole urban network using a Gaussian function. RBF interpolation using the **vehicular flux**, obtained via the discrete Fick's diffusion law, at the nodes of the Athens' road graph (**b**), followed by a diverging colour map, to obtain a smooth representation of incoming and outgoing flows over the entire urban network using a Gaussian function. Colour maps are shared among all vehicles. Both the volume and the flux plots include the visualisation of the gradient vector field, which represents the direction of the largest changes of the corresponding variables and serves as a means to identify **main directions of traffic flows** within an urban network. From the discretisation of Fick's diffusion law, the vehicular flux represents a differential of number of vehicles on a road, and the vehicular volume represents the concentration of vehicles surrounding a road (e.g., at its end points); thus their units can be interpreted as number of vehicles on the road geometry and its boundary, respectively

mobility network, it is crucial to recall that the estimated values are obtained through a linear approach while traffic usually presents more complex trends. Further models that make use of a graph structure in combination with conservation models to solve numerical differential equations on each edge (e.g., the *Cell Transmission Model*) have been applied [6], and a comparison of highlighted critical zones can provide a better understanding of the limitations of the linearity assumption. Similarly, the *METANET* model is another instance of macroscopic models of higher order that have been utilised for urban flow identification [13, 20]. Nevertheless, we consider that using Fick's diffusion law to identify critical zones in a mobility network is meaningful, for instance, for our considered data set based on limited trajectory points, as the other discussed methods were mostly based on Origin-Destination data, turning ratios at intersections, and more detailed network information that requires deeper levels of data.

To assess the adequacy of the discrete Fick's diffusion law as a linear proxy for vehicular flux, we compared its results with a simplified CTM implementation on a representative subnetwork of 120 roads. The mean absolute difference between the two flux estimates was below 8.5%, and the spatial correlation of identified congestion hotspots was $r = 0.89$, with 92% directional agreement in dominant flux orientation.

In addition to this comparison, we extracted quantitative descriptors directly from the reconstructed flux maps (Figs. 3 and 7). First, the magnitude ratio between the strongest accumulation and strongest dispersal zones was $2.7{:}1$, which matches the asymmetric flow behaviour typically captured by CTM on dense urban grids. Second, the spatial footprint of persistent congestion zones (over $t = 0$ to $t + 10$ min) remained concentrated in 11.4% of the network's road length, indicating localised nonlinear effects, but without large-scale propagation where higher-order models would be more determinant. Third, the temporal drift in the dominant congestion centroid across the two timestamps was below 40 m, consistent with slow-moving accumulation that diffusion-based operators capture reliably.

These values show that the linear diffusion model not only preserves the location of critical areas but also approximates their intensity distribution and temporal stability with acceptable accuracy for this dataset. Since the PNeuma trajectories provide snapshot-level measurements rather than full density propagation models (turning ratios, queue spillback rates, or shockwave speeds), higher-order macroscopic models (CTM/METANET) cannot be calibrated meaningfully without additional structural inputs. In this setting, the linear approximation is not a relaxation, but a model/observability match: it captures the dominant macroscopic imbalance signals that are actually supported by the data, without introducing artefacts from unverifiable assumptions.

3.4 Traffic-driven Road Segmentation

Estimating and plotting traffic values on a grid over an urban network represents a solution to obtain large-scale visuals to study traffic. However, they may lead to ambiguous interpretations for areas without any roads within them. To address this matter, we propose the *road segmentation methodology*, consisting of partitioning each road segment along its geometry using a fixed common road distance between consecutive partition nodes and interpolating the reconstructed traffic signals on those nodes. Using this methodology, we plot traffic values only on the road geometries, and it also serves as a means to analyse local changes within a road for different granularity levels, which was not considered when plotting a whole road segment using a single colour.

Since our road network is generated using the OSMNx Python library [3], the geometry of a road is constituted by a *linestring*, which is a list of piecewise linear segments connecting the endpoints of the road. On each road, we use a segmentation algorithm taking as input the original list of linear segments ℓ_i and obtaining as output a new list L of segments defining the same road using the original nodes and the partition nodes. Here, the partition nodes are equally spaced and will be used as the points where traffic values will be estimated through RBF interpolation. Similarly, as before, estimated negative values are mapped to zero. Then, we use heat maps only along the road geometries to visualise the variation of values within the same road, which is particularly noticeable on long roads.

We use the road segmentation methodology with 4, 8, and 16 partitions on each road to assess the visual differences with different granularity levels of the speed values for both RBF interpolation using the reconstructed average speed signal embedded on the road midpoints (Fig. 4). We also analyse the evolution of speed values along incremental times. To this end, we use road segmentation with 20 partitions per road and the traffic state of the roads on three time entries with a difference of 5 minutes. When using the same colour map for every vehicle type and time, besides comparing the speed variations for the vehicle types, we can also visualise the evolution of speed values over time. In this way, we identify the roads where the speed values are decreasing, which could be due to traffic congestion that is being created, and also the roads where speed is increasing, suggesting a traffic decongestion. We again perform our tests with RBF interpolation using the reconstructed average speed signal embedded on the road midpoints (Fig. 5). Finally, we use the road segmentation methodology to visualise the vehicular volume (Fig. 7a) dynamics and their corresponding vehicular flux (Fig. 7b) computed via the discrete Fick's diffusion law for cars and motorcycles at an initial time and 10 minutes later. In this manner, we can see how the congested areas within an urban network are changing over time. Since the vehicular flux signal needs a value defined on each node, before the RBF interpolation at the partition nodes of the road geometries, it is required to reconstruct the volume signal on every network road. We exclude buses because of their considerably lower volume levels and negligible flux values, and evolution. To further evaluate the error in the estimated traffic states obtained through the combined signal recovery and RBF approximation, particularly for average speed, we follow the approach of previous studies [19] but place more emphasis on the visual analysis using the road segmentation method. Specifically, at each partition vertex of the sample roads, we compute the absolute difference between the estimated and original average speed signals and then normalise it by the maximum of the sampled values (Fig. 6). To avoid divisions by zero caused by idle or parked vehicles, we use only the sample nodes with ground-truth values greater than zero. As a result, the number of active roads for buses is much lower, since they are more likely to remain stationary. For both vehicle types, however, the normalised absolute error stays below 20% of the highest ground-truth speed, which is consistent with previous studies. The largest errors occur for motorcycles, especially on road segments in the highest speed zones, suggesting that higher speeds require additional attention in this combined approach, for example, by tuning the

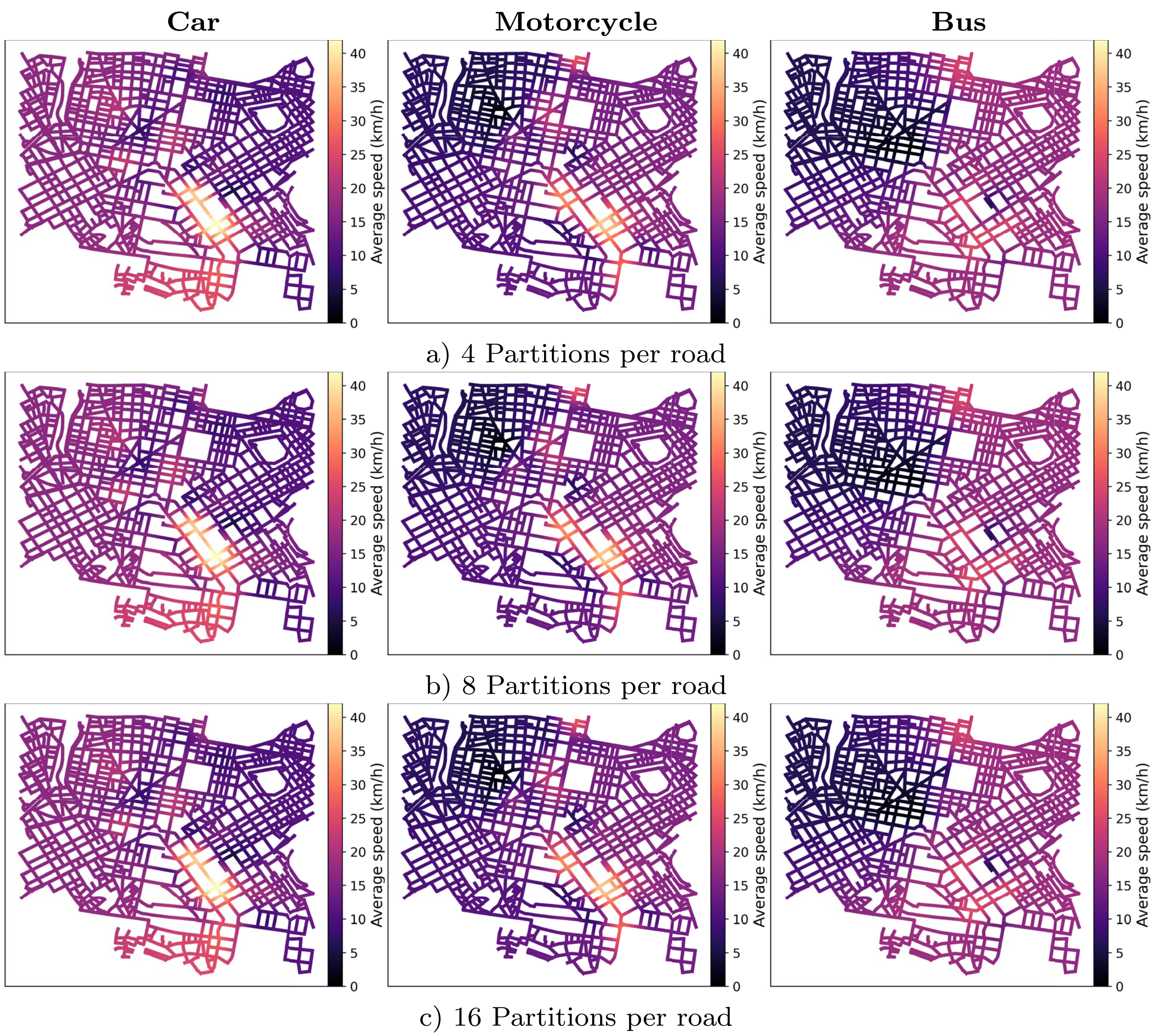

Fig. 4 Road segmentation for different types of vehicles using (**a**) 4, (**b**) 8 and (**c**) 16 partitions per road. The RBF interpolation is done using the **recovered average speed** signal at the midpoint of the roads' geometries. The road segments are colour-coded with heat maps coming from a uniform colour map shared among each type of vehicle

parameters of the graph signal recovery method and the radial basis function.

4 Discussion

The visualisation of the vehicles' positions on the Athens' urban network and the intensity of their instant speeds (Fig. 1) highlight a few reasonable features of traffic dynamics, i.e. a larger number of cars concerning the other vehicle types, a smaller number of buses with also the smallest speeds, and the highest speeds for the motorcycles as it reflects the behaviour of two-wheeled vehicles drivers leveraging the more compact size of their vehicle to navigate within the network. Moreover, the areas with the highest speeds for cars, motorcycles, and taxis are located around the roads with fewer buses. Many vehicles of each type have zero speed, which could stand for parked vehicles, not moving vehicles because of congestion, or vehicles at red traffic lights.

Visualising the RBF interpolation at the vehicles' positions of the instant speeds (Fig. 2a) shows many areas of the urban network with large speeds and a smooth transition to smaller ones. However, those areas with larger values are mainly located outside the roads; therefore, their interpretation can be misleading. In contrast, the RBF interpolation of

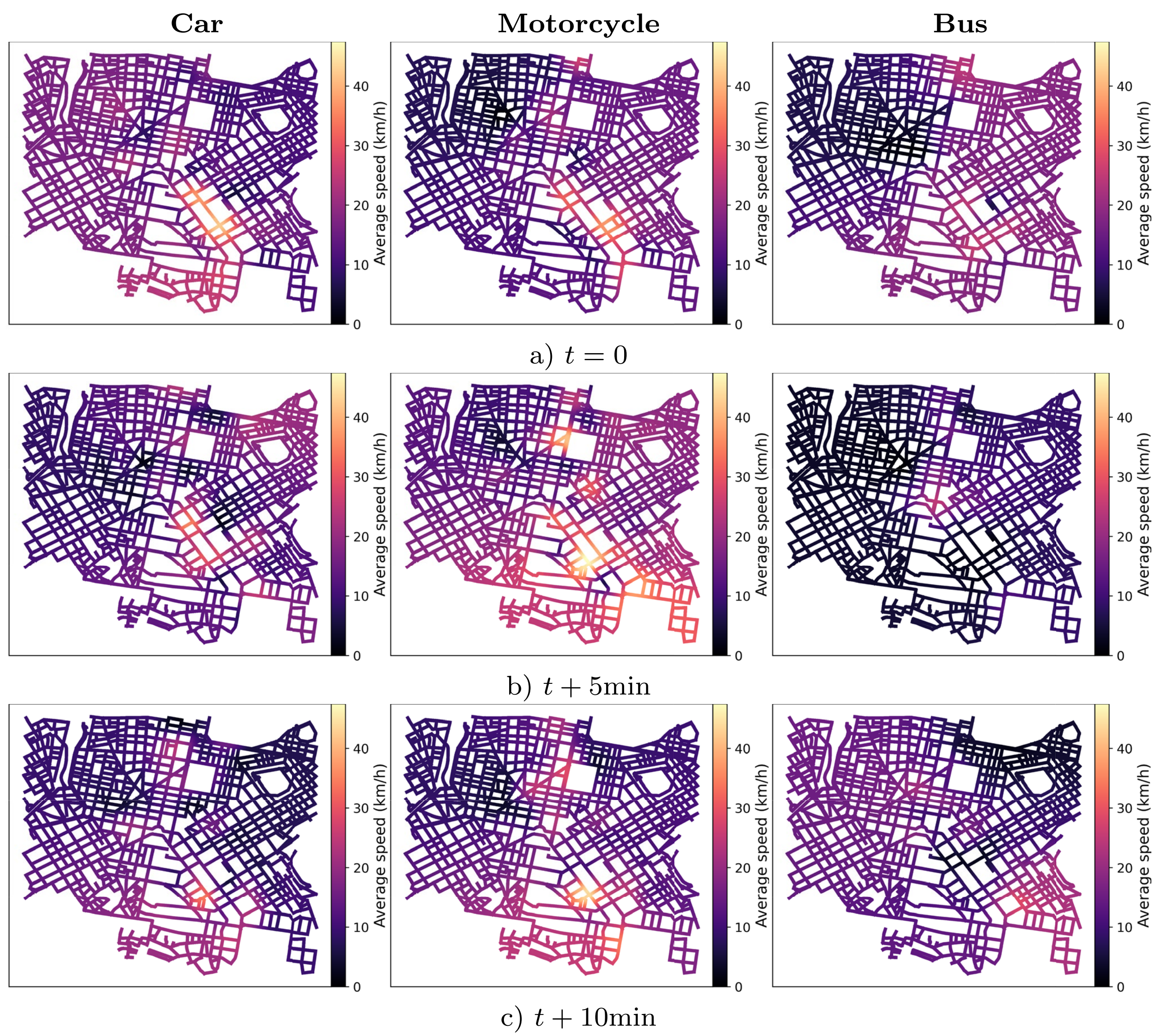

Fig. 5 Road segmentation for different types of vehicles using 20 partitions per road at distinct time entries with a 5min difference; using (**a**) an initial time $t = 0$, (**b**) 5min after t and (**c**) 10min after t. The RBF interpolation is done using the **recovered average speed** graph signal at the midpoint of the roads' geometries. The uniform colour map is shared among each type of vehicle and each time entry

the recovered average speed at the nodes' positions (Fig. 2b) shows considerably fewer areas with larger values, but they are positioned in zones where roads are agglomerating, hence providing a more meaningful estimation. Some other remarkable differences between the two methods are the larger range of values for the interpolation without graph signal recovery and the fewer estimated negative values that are mapped to zero for the graph signal recovery, followed by RBF interpolation.

The visualisation of the RBF interpolation of the recovered volume signal at the midpoint of each road segment (Fig. 3a) highlights the existence of areas with a larger number of vehicles located in similar positions, which is more remarkable for cars and motorcycles. These areas are all located in nearby positions regardless of the type of vehicle, which consistently suggests the existence of potential congestion zones. Moreover, by plotting the gradient vector field of the volume values, we can identify flows within the network that connect zones with small volumes with others that have larger values, which serves as a means to identify main vehicular directions. The number of negative values that are mapped to zero is smaller for cars, in contrast to the motorcycles and buses, and is related to the original number of vehicles before the graph signal recovery process and the number of active roads with at least one projected vehicle.

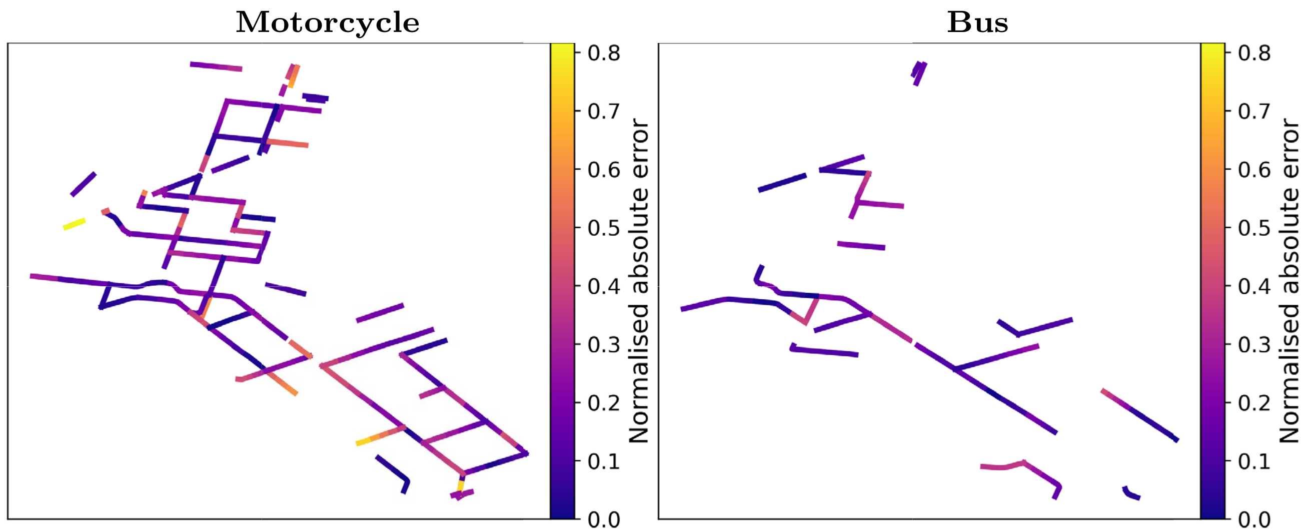

Fig. 6 **Normalised absolute error** of the estimated values using graph signal recovery combined with RBF approximation following the segmentation methodology for the **average speed**, to address the approximation error on sampled roads

Visualising the vehicular flux obtained through the discrete Fick's diffusion law using the recovered volume signal (Fig. 3b) allows the identification of areas of the urban network with more incoming than outgoing vehicles (negative values in red) and vice-versa (positive values in blue). The location of such areas is consistent with one of the highest estimated volumes, as they are geographically close. Estimated vehicular flux complements the analysis of vehicular volume and hints at the existence of zones with traffic congestion, where areas of negative flux with large magnitude are not next to areas of positive flux with large magnitude. For instance, cars have a local maximum of volume at the centre bottom of the urban network; however, its corresponding plot of vehicular flux includes two distinguishable areas with large magnitude (one positive and the other negative), which is interpreted as a temporarily congested area, i.e., it is likely to dissipate earlier than the other congested areas. The inclusion of the orthogonal lines to the level sets enables the visualisation of some flows connecting the critical areas, thus identifying possible congestion and decongestion patterns. The absence of congested areas and critical zones of vehicular flux for buses is interpreted as easier traffic for this type of vehicle without significant obstructions.

Although our approach using RBF approximation can identify critical zones and transitions of values among the urban network by smoothing over noise and signals in the traffic data, it is relevant to remark that sometimes mobility patterns could have abrupt changes. However, for visualisation purposes, based on highlighting zones with minimum and maximum values, our approach is consistent as shown in the different experiments. Moreover, the use of radial basis functions considers interactions among nearby points within a chosen radius, leading to the identification of critical points in the urban network while addressing the changes in their surroundings due to the observable transition of values. In addition, the degree of smoothness can be controlled through the shape ε and regularisation μ parameters, providing a systematic manner to obtain the desired smoothing level, addressing potential over-smoothing caused by the type and distribution of the available data. The choice of an appropriate degree of smoothness and the shape of the radial basis function can be done with existing optimal algorithms to deal with such issues [10, 26].

Regarding the road segmentation methodology, the implementation using 4, 8, or 16 partitions on each road (Fig. 4) shows that the locations of lower and higher speeds are captured regardless of the granularity. The essential difference when varying the number of partitions on each road consists of a smoother variation of values that is more visible on roads of larger lengths. Using road segmentation for the recovered average speed on distinct time entries (Fig. 5) helps to understand the dynamics of vehicles along the network. For instance, the number of roads with higher average speed values changes more considerably for motorcycles than for the other types of vehicles, which can be interpreted as a facility due to their compact sizes and shapes for overcoming traffic impediments that lead to shorter movements. In contrast, the variation of average speeds within roads is less significant for cars as they are not able to easily overcome interruptions of movement. Finally, buses remain with small average speeds along increasing time entries as they have to stop many times to pick up passengers and tend to reduce their velocities more than the other types of vehicles.

Applying the road segmentation methodology to the recovered volume signal and its corresponding vehicular

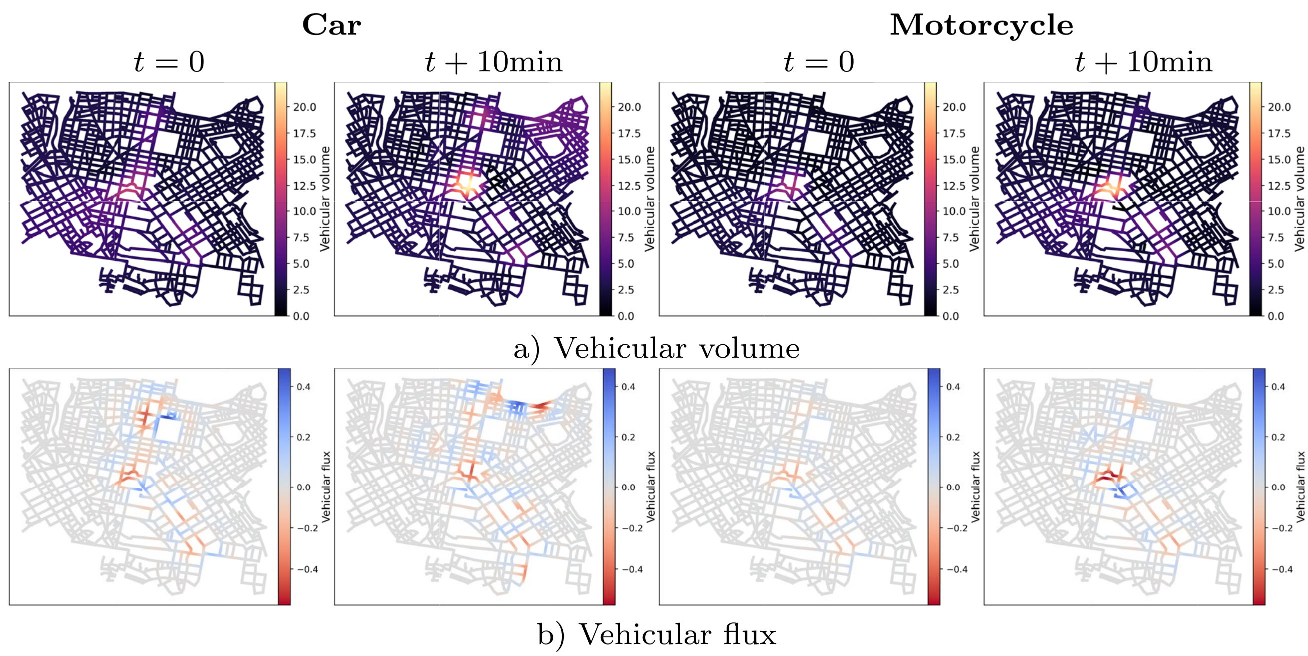

Fig. 7 Road segmentation for cars and motorcycles using 20 partitions per road at two time entries with a 10min difference; using (**a**) **vehicular volume** and (**b**) **vehicular flux** for different types of vehicles using an initial time $t = 0$ and 10min after t. The road segmentation is performed with 20 partitions. The RBF interpolation is done using the recovered graph volume signal at the midpoint of the roads' geometries. The flux is computed via the discrete Fick's diffusion law. The colour map is shared among each type of vehicle and each time entry. From the discretisation of Fick's diffusion law, the vehicular flux represents a differential of number of vehicles on a road, and the vehicular volume represents the concentration of vehicles surrounding a road (e.g., at its end points); thus their units can be interpreted as number of vehicles on the road geometry and its boundary, respectively

flux on two time entries with a distance of 10min for cars and motorcycles, the vehicular volume (Fig. 7a) is higher for the latest time for both types of vehicles, being the central area being the most affected. These plots are consistent with the visualisations using RBF interpolation over a mesh (e.g. Fig. 3). However, when using road segmentation, we can associate traffic estimations directly with the roads of an urban network and also visualise the smooth variation of values within a single road. Using the road segmentation to visualise the vehicular flux computed via the discrete Fick's diffusion law of the corresponding volumes (Fig. 7b), we see again the interdependence between congesting and decongesting areas, i.e., they are neighbours to each other and do not exist independently.

5 Conclusions and Future Work

The visualisation of large amounts of data in the traffic and mobility domains is vital for the identification of dynamic patterns and relevant areas within an urban network. It represents a robust and compelling tool that can be leveraged by citizens, transport providers, and authorities, as it provides a straightforward alternative to understanding the traffic state in a city through a visual illustration. The results of the proposed approach combining RBF interpolation with graph-based recovery methods show that different inferences can be obtained from a single set of traffic data via visual representations. Some remarkable outcomes of the combined approach over a simple RBF interpolation are the reduction of negative estimated values, which is meaningful in the traffic modelling domain, the localisation of critical values over areas with groups of roads instead of empty spaces, and the identification of negative correlated areas for speed and volume, which conveys the idea that large number of vehicles in an area cause the decrease of their speed.

We show that the road segmentation methodology addresses granularity levels within road geometries, which were not directly attained using a typical mesh approach. Moreover, it shows consistent visual results with the ones obtained through mesh-based estimations, which supports its viability. Possible extensions of this work that we plan to address include the impact study of the parameters, i.e. the smoothing term λ of the Laplacian-based graph signal recovery method, the regularisation term μ of the RBF interpolation, and the scaling factor ε of the radial basis functions. Also, partitioning roads with a non-constant number of partitions is an area that merits being explored, for instance, partitioning a road with a number according to some centrality metric since urban networks are a particular case of geographical networks that can leverage this machinery [18]. In future work, it is also relevant to consider the interaction among the distinct types of vehicles. In this work, we aimed to provide tools to visualise the traffic status of distinct types of vehicles within an urban network at various time intervals and levels of detail along the geometry of the roads. Modelling the interaction between the different vehicle categories can contribute to enhancing the understanding of traffic dynamics, and finding a suitable visual representation of such interactions is also valuable in the urban domain. Moreover, the combined visualisation of traffic dynamics patterns with the distribution of points of interest and distinct road categories in a city holds significant potential.

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Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

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